

Waypoint-Based Tracking of a 10 cm Square Trajectory on a Parrot Mambo Minidrone

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Abstract—This report documents trajectory-tracking flights in which a Parrot Mambo minidrone flies a 10 cm square reference twice in succession over an indoor arena. The flight builds on the MathWorks minidrone waypoint-following example, adapted and tuned by the author, with the square encoded as a sequence of position references. At this scale, comparable to the airframe itself, trajectory tracking is unforgiving: corner overshoot and drift are immediately visible against the arena floor. The deployed flight completes both laps in a single take, and the exercise doubles as a repeatability test of the position controller developed in the hover work.

Index Terms—Parrot Mambo, Simulink, waypoint following, trajectory tracking, position control.

I. INTRODUCTION

A hover proves a controller can stand still; a trajectory proves it can go somewhere specific and come back. This flight asks the Mambo to trace a 10 cm square, twice, over the arena floor. The small scale is the point: on a reference comparable to the vehicle's own footprint, every corner is a step change in commanded direction and every controller weakness shows.

The Parrot Mambo carries the sensor suite the whole series relies on: an ultrasound sensor timing pulses to the floor for altitude, a pressure sensor for altitude changes, an IMU providing linear acceleration and angular rates, and a downward-facing camera whose optical flow contributes horizontal velocity estimates. Four motors in an X configuration are commanded through motor mixing: collective thrust, a differential between the clockwise and counter-clockwise pairs for yaw, and side-to-side speed imbalances for pitch and roll. The airframe is underactuated, with six degrees of freedom and only four inputs, so every lateral motion begins as a deliberate tilt of the thrust vector.

II. SYSTEM ARCHITECTURE

All flight code was developed in Simulink against the Parrot Minidrone Support Package and deployed to the physical airframe, keeping the model structure the support package expects: an estimator consuming the sensor buses, a flight control system, and reference logic that the author modified per task. Structuring the work as small reusable subsystems made iteration and debugging on hardware considerably less painful.

The model derives from the minidrone waypoint-follower example project, whose structure the author adapted: the square's

corners become waypoints in the reference logic, and the flight control system tracks the resulting position reference through the cascaded position and attitude loops.

III. METHOD

The square is encoded as a timed sequence of position references, flown at constant altitude, with two full laps commanded back to back. Between corners the position loop drives the drone along each 10 cm edge; at corners the reference direction changes sharply, exercising the controller's transient response eight times per flight.

IV. FLIGHT TESTING AND RESULTS

The deployed flight completes two consecutive laps of the square in one take over the checkerboard arena. Flying the same pattern twice in succession makes lap-to-lap consistency directly visible, and the second lap tracks the same ground region as the first without a reset between laps.



Fig. 1. The Mambo over the checkerboard arena during the square-trajectory flight.

V. CHALLENGES

- Corner transients: a 10 cm edge gives the controller almost no settling distance between step changes.
- Drift accumulating across laps with no external positioning reference indoors.
- Separating controller error from estimation drift when judging the flown path by eye.

VI. CONCLUSION

The flight demonstrates position tracking at a scale where sloppiness cannot hide. Completing the square twice in succession on real hardware validated the position controller for the vision-guided missions that follow in the series.

APPENDIX: MATLAB FUNCTION CODE

The following MATLAB Function blocks are the author's custom code from the deployed Simulink model, reproduced verbatim.

Listing 1. Earth-to-body frame error transformation (flight control model)

```
function xy_err_body = fcn(yaw,xy_err_earth)

transformation = [cos(yaw), sin(yaw); -sin(yaw), cos(yaw)];
xy_err_body = transformation * xy_err_earth;
end
```

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